25901920 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY204 - INTRODUCTION TO MATERIALS SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Write the relation of axes and angles in the rhombohedral system. [U] / [CO1]
2. State one application of Miller indices in materials science. [U] / [CO1]
3. Compare any two properties of ceramics and polymers [U] / [CO1]
4. Schematically plot attractive, repulsive and net force versus interatomic separation for two atoms or ions. Also mark the equilibrium distance [U] / [CO2]
5. What is solvus line? [K] / [CO4]
6. What is zone refining? [K] / [CO4]
7. What is the relationship between transformation temperature and fineness/coarseness of microstructure? [U] / [CO5]
8. What is meant by phase transformation? What is its significance? [U] / [CO5]
9. Define heterogeneous nucleation. Give one example. [K] / [CO5]
10. Why does heterogeneous nucleation occur more readily than homogeneous nucleation? [U] / [CO5]
11. Name two general techniques for improving the oxidation resistance of a metal. [K] / [CO6]
12. Why is ductility of materials important? [U] / [CO6]

13. Why is ductile fracture preferred over brittle fracture? [U] / [CO6]

14. Discuss the concept of thermal fatigue. [U] / [CO6]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Compare ionic and covalent bonding with respect to bond energy, melting point, and conductivity. [U] / [CO1]

16. Draw and explain the (111) plane and the [111] direction in a cubic crystal. [K] / [CO1]

17. Discuss how impurities influence diffusion processes in solids. Support your answer with relevant examples. [U] / [CO3]

18. Comment the following; a) “Phase diagrams provide valuable information about melting, casting, crystallization, and other phenomena.” b) “The addition of solute in excess of the solubility limit results in the formation of another solid solution.” [U] / [CO4]

19. Apply the Gibbs Phase Rule to a binary phase diagram and discuss how it helps in determining the number of degrees of freedom in different regions of the diagram. [U] / [CO4]

20. Explain the different mechanisms of polymer degradation with suitable examples. [U] / [CO6]

21. Explain the stress-strain graph showing nonlinear elastic deformation. [U] / [CO6]

22. Explain the analogy between dislocation motion in crystals and the movement of a caterpillar. [U] / [CO6]

[5x6 = 30]